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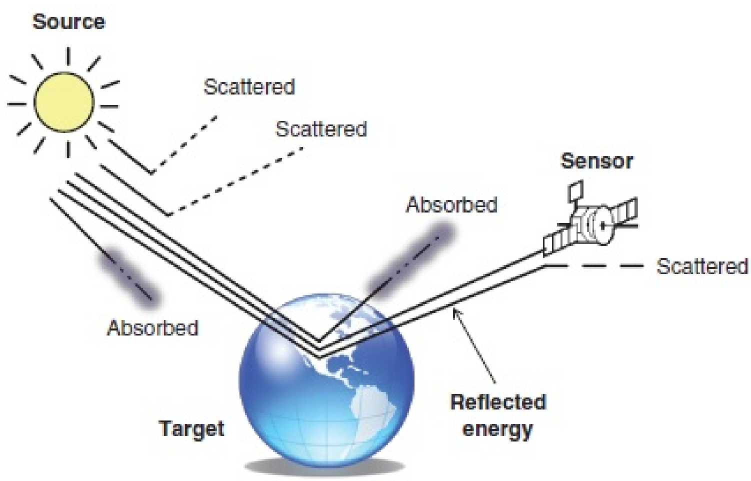
2. Electromagnetic Radiation Principles

2.1. Introduction

The main energy for remote sensing is solar energy, also known as electromagnetic radiation. Photons, the basic unit of electromagnetic radiation, not only move as particles but also as waves of different frequencies and wavelengths. When electromagnetic radiation travels to the Earth and strikes the Earth's surface, it is reflected, transmitted, or absorbed (Fig. 2.1). For any given material or surface, the amount of solar radiation that reflects, absorbs, or transmits varies with wavelength. This important property of matter makes it possible to identify and to separate different substances on the Earth's surface, and forms the basis for remote sensing technology. Before reaching a remote sensor, the electromagnetic radiation has to make at least one journey through the Earth's atmosphere, and two journeys in the case of active systems. Some of the electromagnetic radiation are scattered by atmospheric gases, aerosols, and clouds; while others are absorbed by carbon dioxide, water vapor, or ozone. This chapter first discusses basic properties of major portions of electromagnetic radiation, and then examines how solar radiation interacts with the features on the Earth's surface. This is followed by a detailed discussion of different effects that the atmosphere may generate on electromagnetic radiation and remote sensing images.

Figure 2.1 Solar energy interaction with the environment.



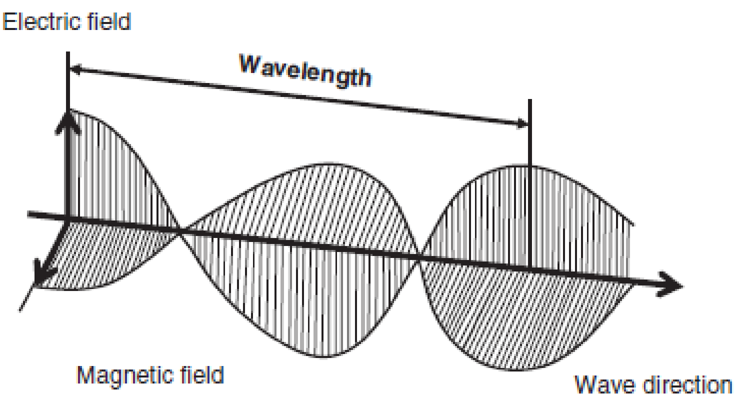
2.2. Principles of Electromagnetic Radiation

Electromagnetic radiation is a form of energy with the properties of a wave, with a major source from the sun. Solar energy traveling in the form of wavelengths at the speed of light (denoted as C, equal to 3×10^8 ms^{-1}) is known as the electromagnetic spectrum. The waves propagate through time and space in a manner rather like water waves, but oscillate in all directions perpendicular to their direction of travel (Fig. 2.2). The fundamental unit of energy for an electromagnetic force is called a photon. The energy E of a photon is proportional to the wave frequency f as expressed in the following equation:

$$E = h f \tag{2.1}$$

where the constant of proportionality h is Planck's Constant (h = 6.626 × 10^{-34} Joules-sec). Radiation from specific parts of the electromagnetic spectrum contains photons of different frequencies and wavelengths. Photons traveling at higher frequencies have more energy. Photons travel at the speed of light, that is, at 3 × 10^8 m per second. Photons not only move as particles but also move as waves, that is, they have a "dual" nature of particles and waves. When a photon travels as an electromagnetic wave, it has two components—one consisting of varying electric fields and the other of varying magnetic fields. Figure 2.2 illustrates this movement. The two components oscillate as sine waves mutually at right angles, and possess the same amplitudes (strengths), reaching their maxima-minima at the same time. Unlike other types of waves that require a carrier (e.g., water waves), photon waves can transmit through a vacuum (such as in space). When photons pass from one medium to another, for example, air to glass, their wave pathways are bent (follow new directions) and thus experience refraction (Short, 2009).

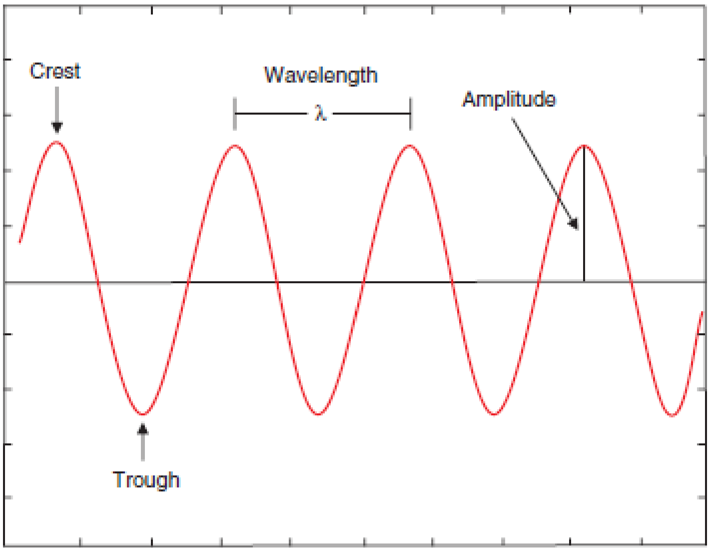
Figure 2.2 Electromagnetic radiation travels in the form of a wave.



When any material on the Earth's surface interacts with incoming electromagnetic radiation, or is excited by internal processes, it will reflect, or emit, the amount of photons that differ at different wavelengths. Photon energy received by a remote sensor is commonly stated in power units such as Watts per square meter per wavelength unit. The plot of variation of power with wavelength gives rise to a specific pattern that is diagnostic of the material being sensed.

Electromagnetic waves are commonly characterized by two measures: wavelength and frequency. The wavelength (λ) is the distance between successive crests of the waves. The frequency (μ) is the number of oscillations completed per second. Other terms are also needed in order to better describe electromagnetic waves, including crest, trough, amplitude, and period (Fig. 2.3). Crest is the highest point of a wave, while trough is the lowest point. Amplitude is another important term in the wave description, referring to the distance from a wave's midpoint to its crest or trough. Period is the time it takes for two consecutive crests to pass a stationary point.

Figure 2.3 *Key terms used for describing electromagnetic waves.*

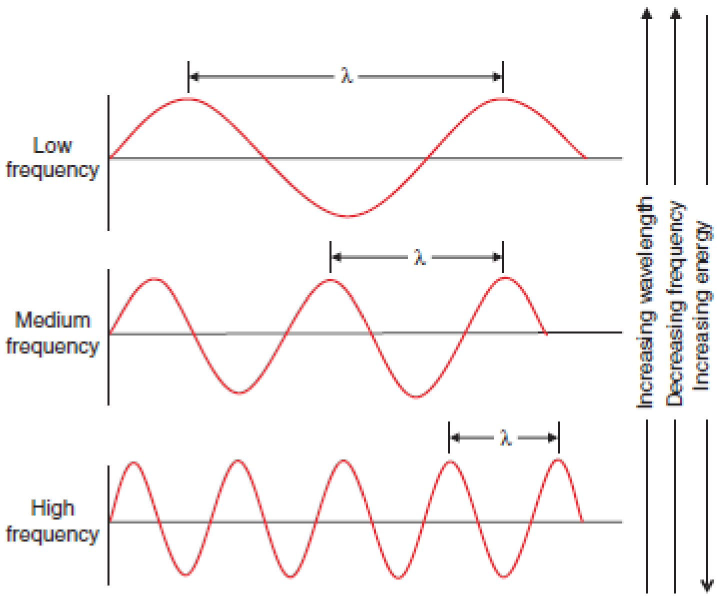


Wavelength and frequency are related by the following equation:

$$C = \lambda^{*}\mu \quad (2.2)$$

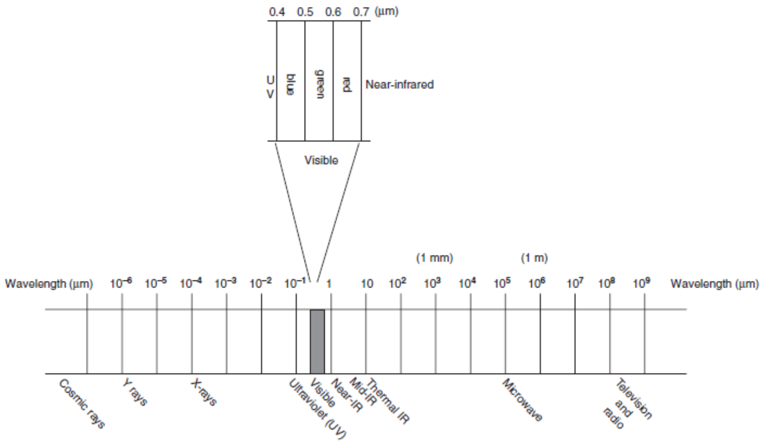
Because the speed of light, *C*, is a constant, there exists an inverse relationship between the wavelength and the frequency of electromagnetic waves. As the wavelength becomes longer, the frequency decreases, and vice versa. In other words, a high-frequency electromagnetic wave (i.e., high-energy wave) has a shorter wavelength. Figure 2.4 further illustrates this inverse relationship between wavelength and frequency.

Figure 2.4 *Inverse relationship between wavelength and frequency.*



The electromagnetic spectrum, despite being thought of as a continuum of wavelengths and frequencies, may be divided into different portions by scientific conventions (Fig. 2.5). Major divisions of the electromagnetic spectrum, ranging from short wavelength, high frequency waves to long wavelength, low frequency waves, include gamma rays, X-rays, ultraviolet, visible, infrared spectrum, microwave, and radio wave. Therefore, in comparison of X-rays and microwave, X-rays have much shorter wavelength, higher frequency, and, therefore, higher energy.

Figure 2.5 Major divisions of the electromagnetic radiation spectrum.



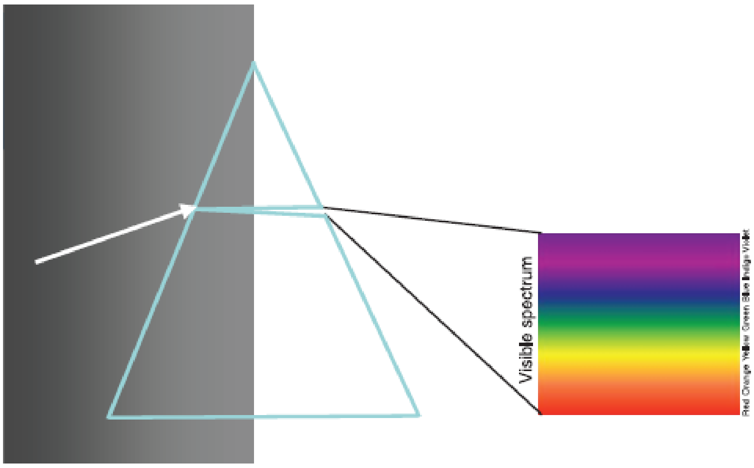
Among the major divisions of the electromagnetic spectrum, gamma rays have the shortest wavelength and highest frequency. Their wavelengths are less than approximately 10 trillionths of a meter, and are extremely penetrating. Gamma rays are generated by radioactive atoms and in nuclear explosions. These rays have been used in many medical, as well as astronomical applications. For example, images of our universe taken in gamma rays have yielded important information on the life and the death of stars and other violent processes in the universe. Gamma rays are not used in remote sensing because they are completely absorbed by the upper atmosphere in the process of traveling to the Earth (Sabins, 1997).

We are familiar with X-rays due to their wide use in the medical field. X-rays have wavelengths in the range from approximately 10 billionths of a meter to approximately 10 trillionths of a meter. Like gamma rays, X-rays are high-energy waves with great penetrating power. In addition to the medical field, they are also extensively utilized in weld inspections and astronomical observations. X-ray images of the sun provide important information on changes in solar flares that can affect space weather. X-rays from the sun are absorbed by the atmosphere before reaching the Earth, and therefore are not used in remote sensing.

Ultraviolet radiation has a range of wavelengths from 400 billionths of a meter (0.4 μm) to approximately 10 billionths of a meter (0.01 μm). Solar radiation contains ultraviolet waves, but most of them are absorbed by ozone in the Earth's upper atmosphere. A small dose of ultraviolet radiation that reaches the Earth's surface is beneficial to human health. Because larger doses of ultraviolet radiation can cause skin cancer and cataracts, scientists are concerned about the measurement of ozone and stability of the ozone layer. Only a small portion of ultraviolet radiation (in the range of 0.3 to 0.4 μm) can reach the Earth's surface, and can potentially be used for remote sensing of terrestrial features. However, this portion of ultraviolet radiation is easily scattered by the atmosphere owing to its short wavelength. This atmospheric effect reduces the usefulness of ultraviolet radiation in remote sensing of Earth materials (Campbell, 2007). Ultraviolet wavelengths are also used extensively in astronomical observatories.

The visible spectrum, commonly known as "sunlight," is the portion of the electromagnetic spectrum with wavelengths between 400 and 700 billionths of a meter (0.4 to 0.7 μm). This range is established by the sensitivity of human eyes to electromagnetic radiation. The majority of solar radiation reaches the top of the Earth's atmosphere as white light, which is separated into familiar rainbow colors when passing through a glass prism (Fig. 2.6). In remote sensing, however, the visible spectrum is usually divided into three segments: blue (0.4 to 0.5 μm), green (0.5 to 0.6 μm), and red (0.6 to 0.7 μm). The three segments refer to the additive primary colors, with which the mixture of certain proportions of blue, green, and red can form all other colors. Approximately 40% of solar energy is concentrated in the visible spectrum, and it is relatively easy to go through the atmosphere reaching the Earth's surface (Avery and Berlin, 1992). Colors we see in nature are the result of reflection from particular objects. The maximum amount of electromagnetic radiation reflected from the Earth's surface is at the wavelength of 0.5 μm, corresponding to the green wavelength of the visible spectrum (Sabins, 1997). Therefore, the visible spectrum is very suitable for remote sensing of vegetation and for the identification of different objects by their visible colors in photography.

Figure 2.6 Visible spectrum when passing through a glass prism.



The infrared spectrum is the region of electromagnetic radiation that extends from the visible region to approximately 1 mm (in wavelength). Infrared waves can be further partitioned into near-infrared (0.7 to 1.3 μm), mid-infrared (1.3 to 3 μm), and far-infrared spectrum (3 to 1000 μm). However, due to atmospheric absorption and scattering, infrared radiation beyond 14 μm is not available for remote sensing. The infrared spectrum between 3 and 14 μm is also known as thermal radiation, or emitted radiation, and in plain language, "heat." In remote sensing, the near- and mid-infrared spectrums are conventionally combined to be called reflected infrared band, which acts similarly as the visible spectrum. The spectral property of reflected infrared radiation is very sensitive to vegetation vigor. Therefore, infrared images obtained by thermal sensors can yield important information on the health of crops and other types of vegetations. Solar energy contains a tiny amount of thermal infrared radiation. The thermal energy we use in remote sensing is mainly emitted by the Earth and the atmosphere. Thermal infrared radiation can help us monitor volcano activity, detect heat leaks from our houses, and see forest fires even when they are enveloped in an opaque curtain of smoke. Because the Earth emits thermal radiation day and night with the maximum energy radiating at the 9.7 μm wavelength (Sabins, 1997), some remote sensors have been designed to detect energy in the range from 8 to 12 μm in remote sensing of terrestrial features.

Microwave radiation has a wavelength ranging from approximately 1 mm to 1 m. Microwaves are emitted from Earth and the atmosphere, and from objects such as cars and planes. These microwaves can be detected to give information, such as on the temperature of an object that emits the microwave. Because their wavelengths are so long, the energy available is quite small compared to visible and infrared wavelengths. Therefore, the field of view must be large enough to detect sufficient energy in order to record a signal. Most passive microwave sensors are thus characterized by low spatial resolution. Active microwave sensing systems (such as radar) provide their own source of microwave radiation to illuminate the targets on the ground. A major advantage of radar is the capability of the radiation to penetrate through cloud cover and most weather conditions owing to its long wavelength. In addition, because radar is an active sensor, it can also be used to image the ground at any time during the day or night. These two primary advantages of radar, that is, all-weather and day or night imaging, make it a unique sensor.

Radio waves have wavelengths that range from approximately 30 cm to tens or even hundreds of meters. Radio waves can generally be utilized by antennas of appropriate size (according to the principle of resonance) for transmission of data via modulation. Television, mobile phones, wireless networking, and amateur radio all use radio waves. Radio waves can be made to carry data by varying a combination of the amplitude, frequency, and phase of the wave within a frequency band. The reflected radio waves can also be used to form an image of the ground features in complete darkness or through clouds.

2.3. Interactions with Earth Surface Materials

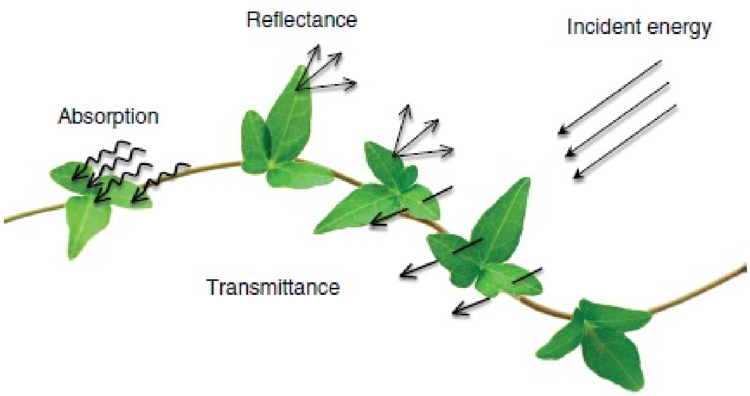
The electromagnetic radiation that impinges on the Earth's surface is called incident radiation. By interacting with features on the Earth's surface, the electromagnetic radiation is partitioned into three types, that is, reflection, transmission, and absorption. The proportion of the electromagnetic radiation that each type of interaction receives depends mainly on three factors: (1) the compositional, biophysical, and chemical properties of the surface feature/material; (2) the wavelength, or frequency, of the incident radiation; and (3) the angle at which the incident radiation strikes the surface (Avery and Berlin, 1992). Reflection is the bouncing of electromagnetic energy from a surface. Reflectance is the term to define the ratio of the amount of electromagnetic radiation reflected from a surface to the amount originally striking the surface. Transmission refers to the movement of energy through a surface. The amount of transmitted energy is wavelength dependent, and is measured as the ratio of transmitted radiation to the incident radiation known as transmittance. Some electromagnetic radiation is absorbed through electron or molecular reactions within the medium. Absorptance is the term to define the ratio between the amount of radiation absorbed by the medium (surface feature or material) and the incident radiation. According to the principle of energy conservation, we can establish the following two equations:

$$\text{Incident radiation} = \text{reflected radiation} + \text{transmitted radiation} + \text{absorbed radiation} \tag{2.3}$$

$$\text{Reflectance} + \text{Transmittance} + \text{Absorptance} = 1 \tag{2.4}$$

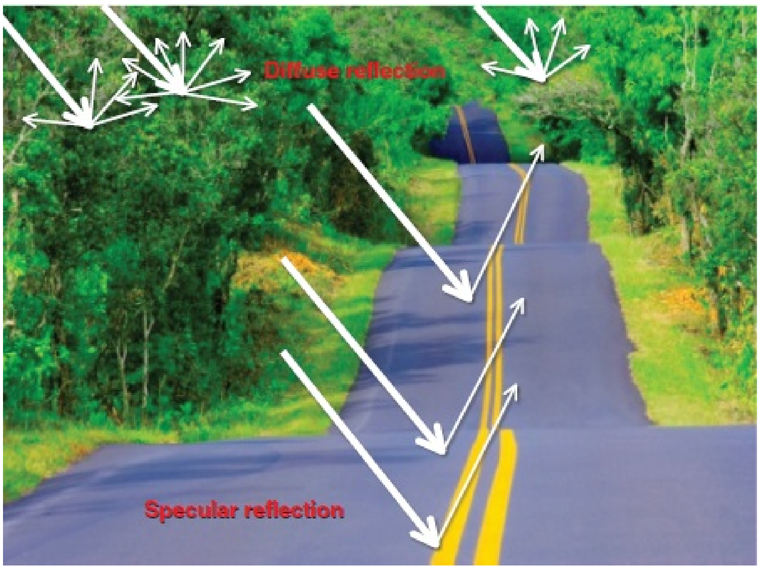
A portion of the absorbed energy is then re-emitted as emittance, usually at longer wavelengths, and the other portion remains and heats the target. Figure 2.7 illustrates the three types of interaction between electromagnetic radiation and the Earth's surface features using a leaf as an example.

Figure 2.7 *Three types of interaction between electromagnetic radiation and leaves.*



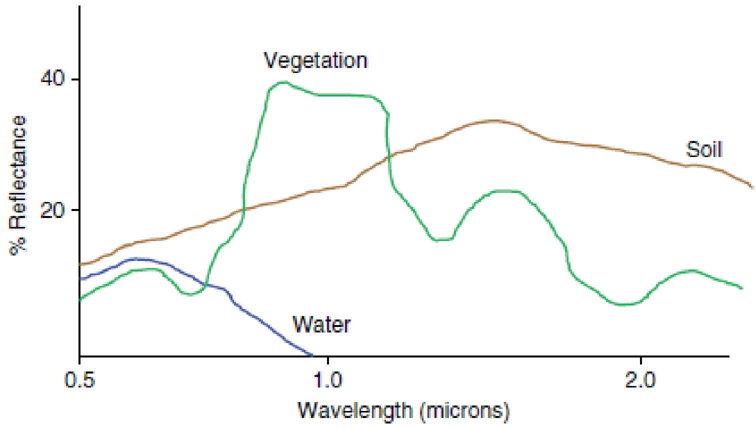
When a surface is smooth, we get specular reflection where all (or almost all) of the energy is directed away from the surface in a single direction (Fig. 2.8). Specular reflectance within the visible wavelength ranges from as high as 0.99 for a very good mirror to as low as 0.02 to 0.04 for a very smooth water surface (Short, 2009).When the surface is rough and the energy is reflected almost uniformly in all directions, diffuse reflection occurs (Fig. 2.8). Most of the Earth's surface lies somewhere between a perfectly specular and a perfectly diffuse reflector. Whether a particular target reflects specularly or diffusely, or somewhere in-between, depends on the roughness of the surface in comparison to the wavelength of the incident radiation. If the wavelengths are much smaller than the surface variations or the particle sizes that make up the surface, diffuse reflection will dominate.

Figure 2.8 *Specular and diffuse reflection.*



Remote sensing systems can detect and record both reflected and emitted energy from the Earth's surface by using different types of sensors. For any given material on the Earth's surface, the amount of solar radiation that reflects, absorbs, or transmits varies with wavelength. Theoretically speaking, no two materials would have the same amount of reflectance, absorption, and transmission within a specific range of wavelength. This important property of matter makes it possible to identify different substances or features and separate them by their "spectral signatures." When reflectance, absorption, or transmission is plotted against wavelength, a unique spectral curve is formed for a specific material. Figure 2.9 illustrates the typical spectral (reflectance) curves for three major terrestrial features—vegetation, water, and soil. Using their reflectance differences, we can distinguish these common Earth surface materials. When using more than two wavelengths, the plot in a multidimensional space can show more separation among the materials. This improved ability to distinguish materials due to extra wavelengths is the basis for multi-spectral remote sensing.

Figure 2.9 Typical spectral curves for three major terrestrial features—vegetation, water, and soil.



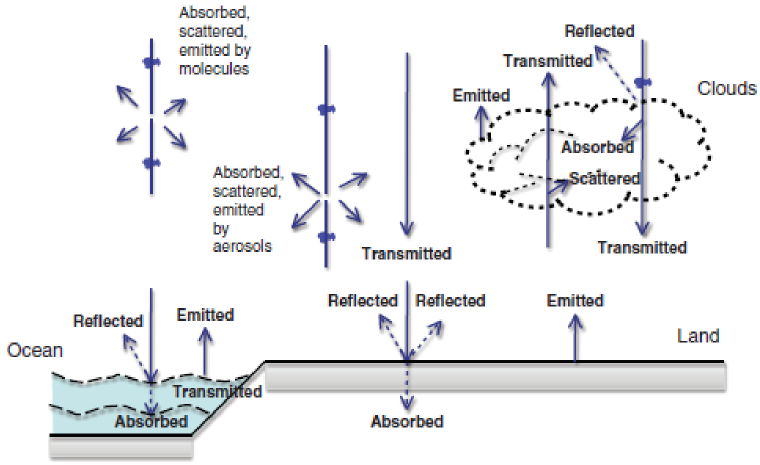
2.4. Atmospheric Effects

Radiation detected by satellite remote sensors passes through the atmosphere with some path length. The atmosphere affects electromagnetic energy through the processes of absorption, scattering, and reflection. How much these processes affect the radiation along the path to the satellite depends on the path length, the presence of particulates and absorbing gases, and the wavelengths involved. Figure 2.10 illustrates how electromagnetic radiation is partitioned in the atmosphere.

Scattering is mainly caused by N_2, O_2 molecules, aerosols, fog particles, cloud droplets, and raindrops. Electromagnetic radiation is lost by being redirected out of the beam of radiation, but the wavelength does not change. Atmospheric scattering may impact remote sensing and resulted images in different ways, including causing skylight (allowing us to see in shadow), forcing images to record the brightness of the atmosphere in addition to the target on the Earth's surface, directing reflected light away from the sensor aperture or light normally outside the sensor's field of view toward the sensor's aperture (both decreasing the sharpness of the images), and making dark objects lighter and light objects darker (thus reducing the contrast).

Atmospheric scattering is accomplished through absorption and immediate re-emission of radiation by atoms or molecules. Rayleigh scattering occurs when radiation interacts with particles or molecules much smaller in diameter than the wavelength of the radiation. This type of scattering dominates in the upper 4 to 5 km of the atmosphere, but can take place up to 9 to 10 km above the Earth's surface. The degree of Rayleigh scattering is inversely proportional to the fourth power of the wavelength. As a result, short wavelength radiation tends to be scattered much more than long-wave radiation. The wavelength dependence and lack of directional dependence of Rayleigh scattering result in the appearance of "blue" sky.

Figure 2.10 Process of atmospheric radiation (After LaDue, J. and K. Pryor, 2003. NOAA/NESDIS).



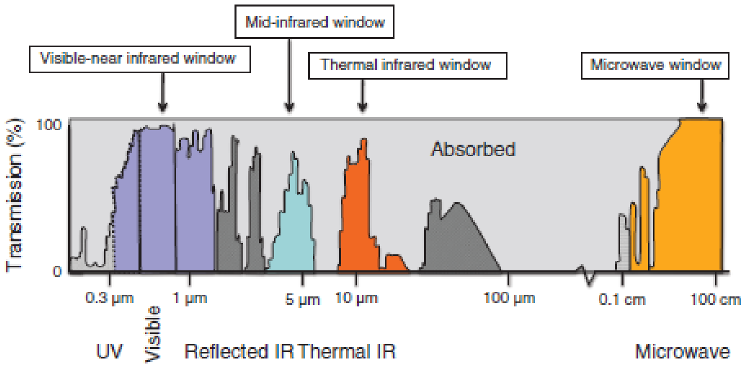
If the atmospheric particulates are of roughly the same size as the wavelength of incident radiation, Mie scattering becomes important. This type of scattering takes place in the lower atmosphere (0 to 5 km) and can affect longer wavelength radiation. Smoke, dust, pollen, and water droplets are the dominant sources of Mie scattering. The orange and red light visible in the sky during sunrise and sunset are the result of this type of scattering. The greater the amount of smoke and dust particles in the atmosphere, the more that violet and blue light will be scattered away and only the longer orange and red wavelength light will be visible. Unlike Rayleigh scattering, most of the incident radiation is scattered in a forward direction because of Mie scattering. Mie scattering is the primary cause of haze, which often scatters sunlight into the sensor's field of view, degrading the quality of remote sensing images.

Non-selective scattering takes place in the lowest portions of the atmosphere where there are particles 10 times greater than the wavelength of the incident electromagnetic radiation. All wavelengths of light are scattered. The water droplets and ice crystals that comprise clouds scatter all visible light wavelengths equally well. Clouds appear white because of this type of scattering.

Atmospheric absorption is the process in which the electromagnetic radiation is absorbed and re-radiated again in all directions, and probably over a different range of wavelengths. Absorption can occur due to atmospheric gases, aerosols, clouds, and precipitation particles, but three types of atmospheric gasses mostly cause it—ozone, carbon dioxide, and water vapor. Ozone absorbs ultraviolet radiation, while carbon dioxide in the lower atmosphere absorbs energy in the spectral region of 13 to 17.5 μm. Water vapor in the lower atmosphere can be very effective in absorbing the energy in the spectrum between 5.5 and 7 μm and above 27 μm. The absorption by water vapor is an important consideration in remote sensing of humid regions. Overall, certain wavelengths of radiation are affected more by absorption than by scattering, especially in the infrared spectrum and wavelengths shorter than the visible portion of the electromagnetic spectrum. Where atmospheric absorption is high, the atmosphere must emit more radiation because of thermal excitation. Within these spectral regions, the surface of the Earth is not visible from space. However, these wavelengths are useful for observing (remote sensing) the atmosphere.

However, there are certain portions of the electromagnetic spectrum that can pass through the atmosphere with little or no attenuation (scattering and absorption). These spectral regions are called atmospheric windows. Four principal windows (by wavelength interval) open to effective remote sensing from above the atmosphere include: (1) visible-near infrared (0.4 to 2.5 μm); (2) mid-infrared (3 to 5 μm); (3) thermal infrared (8 to 14 μm); and (4) microwave (1 to 30 cm). Figure 2.11 shows these four atmospheric windows.

Figure 2.11 Atmospheric windows.



2.5. Summary

The electromagnetic spectrum may be divided into different portions based on wavelength and frequency, including gamma rays, X-rays, ultraviolet, visible, infrared spectrum, microwave, and radio wave. Electromagnetic radiation interacts with the Earth's surface features by reflection, absorption, or transmission. The amount of reflectance, absorption, and transmittance from a specific substance or material vary with wavelengths. The basic principle of remote sensing is recording the reflectance, in some case emittance because of absorption, by a remote sensor, and processing and interpreting acquired images in subsequent steps. Subsequent chapters of this book will describe the methods and techniques for processing and interpreting remote sensing images and data.

The Earth's atmosphere has major effects on remote sensing via scattering and absorption. In either case, the energy is attenuated in the original direction of the radiation's propagation. Therefore, the portions of the electromagnetic spectrum that can pass through the atmosphere with little or no attenuation are critical for remote sensing of Earth surface features. These atmospheric windows lie in the spectrum from visible to infrared and microwave.

The remote sensing principle described in this chapter is applied mainly to passive remote sensing systems, which record the reflected energy of the electromagnetic radiation or the emitted energy from the Earth by using cameras and thermal infrared detectors. Remote sensing can take another form via active sensing systems. The examples of active remote sensing include radar and Lidar. These remote sensing systems send out their own energy and record the reflected portion of that energy from the Earth's surface, and thus they can be used at daytime or nighttime and under almost all weather conditions. **Chapter 10** focuses on active remote sensing.

2.6. Key Concepts and Terms

Absorptance The ratio between the amount of radiation absorbed by the medium (surface feature or material) and the incident radiation.

Absorption The portion of the electromagnetic radiation that is absorbed through electron or molecular reactions within the medium.

Atmospheric absorption The process in which the electromagnetic radiation is absorbed by atmospheric gases, aerosols, clouds, and precipitation particles and re-radiated again in all directions, and probably over a different range of wavelengths.

Atmospheric scattering Scattering is mainly caused by N_2, O_2 molecules, aerosols, fog particles, cloud droplets, and raindrops. Electromagnetic radiation is lost by being redirected out of the beam of radiation, but wavelength does not change.

Atmospheric window The portion of the electromagnetic spectrum that transmits radiant energy effectively.

Diffuse reflection Occurs when the surface is rough. The energy is reflected almost uniformly in all directions.

Electromagnetic radiation A form of energy with the properties of a wave. The waves propagate through time and space in a manner rather like water waves, but oscillate in all directions perpendicular to their direction of travel.

Electromagnetic spectrum The sun's energy travels in the form of wavelengths at the speed of light to reach to the Earth's surface. The energy is composed of several different wavelengths, and the full range of wavelengths is known as the electromagnetic spectrum.

Frequency The number of oscillations completed per second.

Gamma rays Wavelengths of gamma rays are less than approximately 10 trillionths of a meter. Gamma rays are generated by radioactive atoms and in nuclear explosions, and are used in many medical and astronomical applications.

Incident radiation Electromagnetic radiation that interacts with an object on the Earth's surface.

Infrared spectrum The region of the electromagnetic spectrum that extends from the visible region to approximately 1 mm in wavelength. Infrared radiation can be measured using electronic detectors, and has applications in medicine and in detecting heat leaks from houses, crop health, forest fires, and in many other areas.

Microwave radiation Wavelengths range from approximately 1 mm to 1 m. Microwaves are emitted from the Earth, from objects such as cars and planes, and from the atmosphere. These microwaves can be detected to give information, such as the temperature of the object that emitted the microwaves.

Mie scattering If the atmospheric particulates are of roughly the same size as the wavelength of incident radiation, Mie scattering becomes important. This type of scattering takes place in the lower atmosphere (0 to 5 km) and can affect longer wavelength radiation. Smoke, dust, pollen, and water droplets are the dominant sources of Mie scattering.

Non-selective scattering Takes place in the lowest portions of the atmosphere where there are particles 10 times greater than the wavelength of the incident electromagnetic radiation. All wavelengths of light are scattered. The water droplets and ice crystals that comprise clouds scatter all visible light wavelengths equally well. Clouds appear white because of this type of scattering.

Photon The fundamental unit of energy for an electromagnetic force.

Radio waves Wavelengths range from less than 1 cm to tens or even hundreds of meters. Radio waves are used to transmit radio and television signals. The reflected waves can be used to create images of the ground in complete darkness or through clouds.

Rayleigh scattering Occurs when radiation interacts with particles or molecules much smaller in diameter than the wavelength of the radiation. This type of scattering dominates in the upper 4 to 5 km of the atmosphere, but can take place in up to 9 to 10 km above the Earth's surface.

Reflectance The proportional amount of the electromagnetic radiation reflected from a surface to the amount originally striking the surface.

Reflection The bouncing of electromagnetic energy from a surface.

Spectral signatures For any given material, the amount of solar radiation that reflects, absorbs, or transmits varies with wavelength. This important property of matter makes it possible to identify different substances or classes and separate them by their spectral signatures (spectral curves).

Specular reflection Occurs when a surface is smooth. All (or almost all) of the energy is directed away from the surface in a single direction.

Transmission The movement of light through a surface. Transmission is wavelength dependent.

Transmittance The proportional amount of incident radiation passing through a surface. It is measured as the ratio of transmitted radiation to the incident radiation.

Ultraviolet radiation The wavelength ranges from 400 billionths of a meter to approximately 10 billionths of a meter. Sunlight contains ultraviolet waves that can burn human's skin. Ultraviolet wavelengths are used extensively in astronomical observatories.

Visible spectrum A portion of the electromagnetic spectrum with wavelengths between 400 and 700 billionths of a meter (400 to 700 nm). The rainbow of colors we see is the visible spectrum.

Wavelength The distance between successive crests of waves.

X-rays High-energy waves that have great penetrating power and are used extensively in medical and astronomical applications and in inspecting welds. The wavelength range is from approximately 10 billionths of a meter to approximately 10 trillionths of a meter.

2.7. Review Questions

1. How does electromagnetic radiation interact with an object on the Earth's surface, for example, with a tree? Explain it by identifying different types of interactions.

2. Human beings cannot "see" things in the dark, but some satellite sensors can. Explain why.

3. Among the three materials—vegetation, soil, and water—which materials have the highest and lowest reflectance in the visible spectrum (0.4 to 0.7 μm)? In which region of the spectrum—visible, near-infrared (0.7 to 1.3 μm), or mid-infrared (1.3 to 3 μm)—would the three materials be most easily separated?

4. In order to detect heat leaks from houses, or potential forest fires, which region of the spectrum should be used? Which regions of the spectrum have been used in astronomical observations?

5. What effect will atmospheric scattering have on satellite images? Please give an example of Rayleigh, Mie, and non-selective scattering.

6. What are atmospheric windows? How do they affect remote sensing?

7. The Amazon rain forest is the largest tropical jungle in the world. Clear cutting in this region has become a major environmental concern because it destructs habitats and disturbs the balance between the oxygen and carbon dioxide produced and used in photosynthesis. What would be some major difficulties for remote sensing of tropical forests there?

2.8. References

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